

6. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0S1 \mid \epsilon$

Code:

```
#include <stdio.h>
#include <string.h>
int main() {
    char s[100];
    int i = 0, c0 = 0, c1 = 0;

    printf("Enter string: ");
    scanf("%s", s);

    while (s[i] == '0') {
        c0++;
        i++;
    }

    while (s[i] == '1') {
        c1++;
        i++;
    }

    if (s[i] == '\0' && c0 == c1)
        printf("String belongs to CFG");
    else
        printf("String does not belong to CFG");
    return 0;
}
```

Output:

<pre>#include <stdio.h> #include <string.h> int main() { char s[100]; int i = 0, c0 = 0, c1 = 0; printf("Enter string: "); scanf("%s", s); while (s[i] == '0') { c0++; i++; } while (s[i] == '1') { c1++; i++; } if (s[i] == '\0' && c0 == c1) printf("String belongs to CFG"); else printf("String does not belong to CFG"); return 0; }</pre>	<pre>Enter string: 0011 String belongs to CFG === Code Execution Successful ===</pre>
---	--

